# ASC | ICAM



# ASCII IMAGE &VIDEO CONVERTING SYSTEM



# PROBLEM STATEMENT



Converting images and videos into ASCII art often requires complex tools or manual effort. This project aims to provide a tool that converts images and videos into ASCII art quickly and efficiently, directly from digital media.

# SOLUTION

**Develop a user-friendly web tool that converts images and videos into ASCII art with customizable characters, resolution, and downloadable output.**





# KEY FEATURES

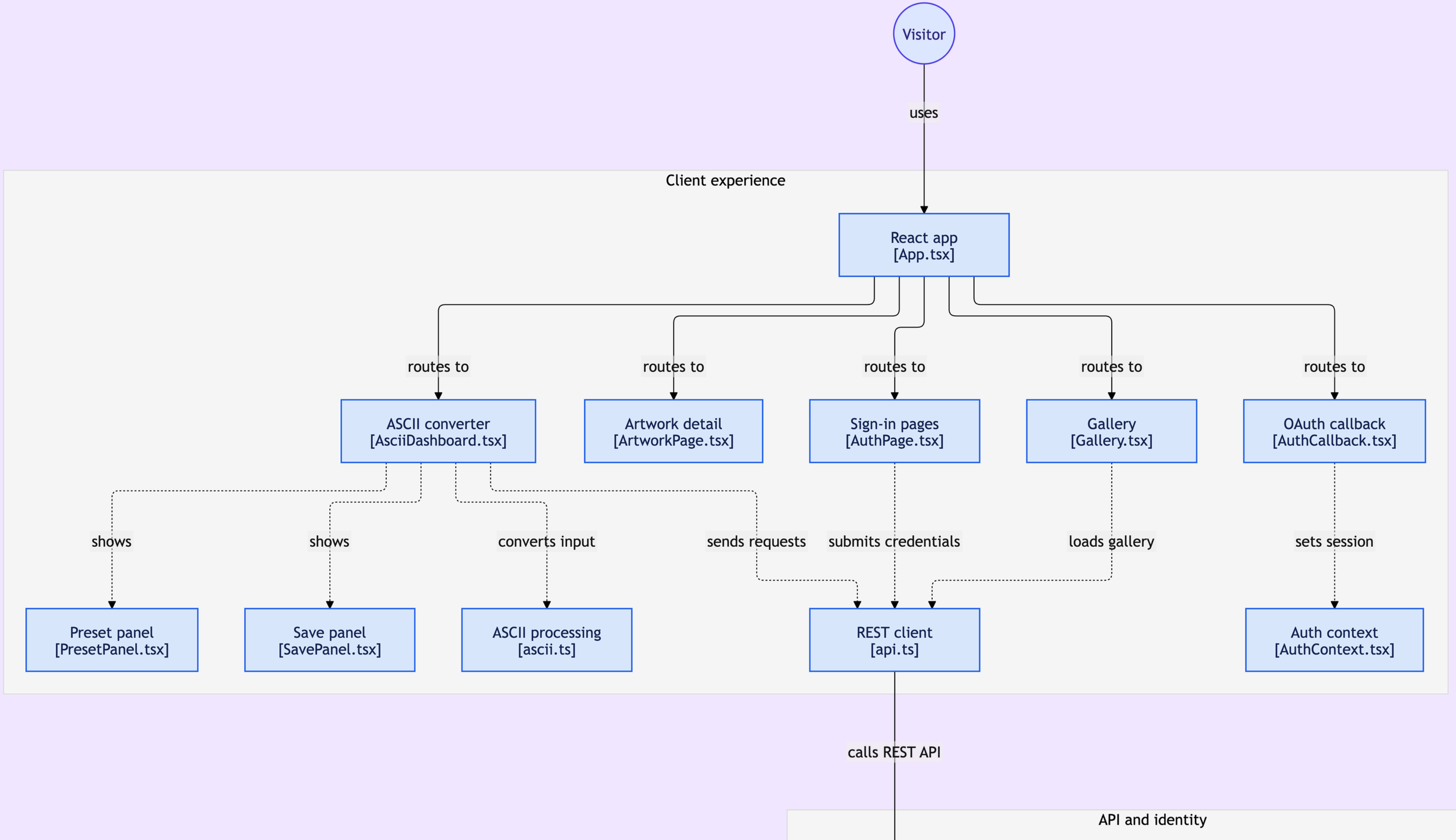
**Main capabilities**

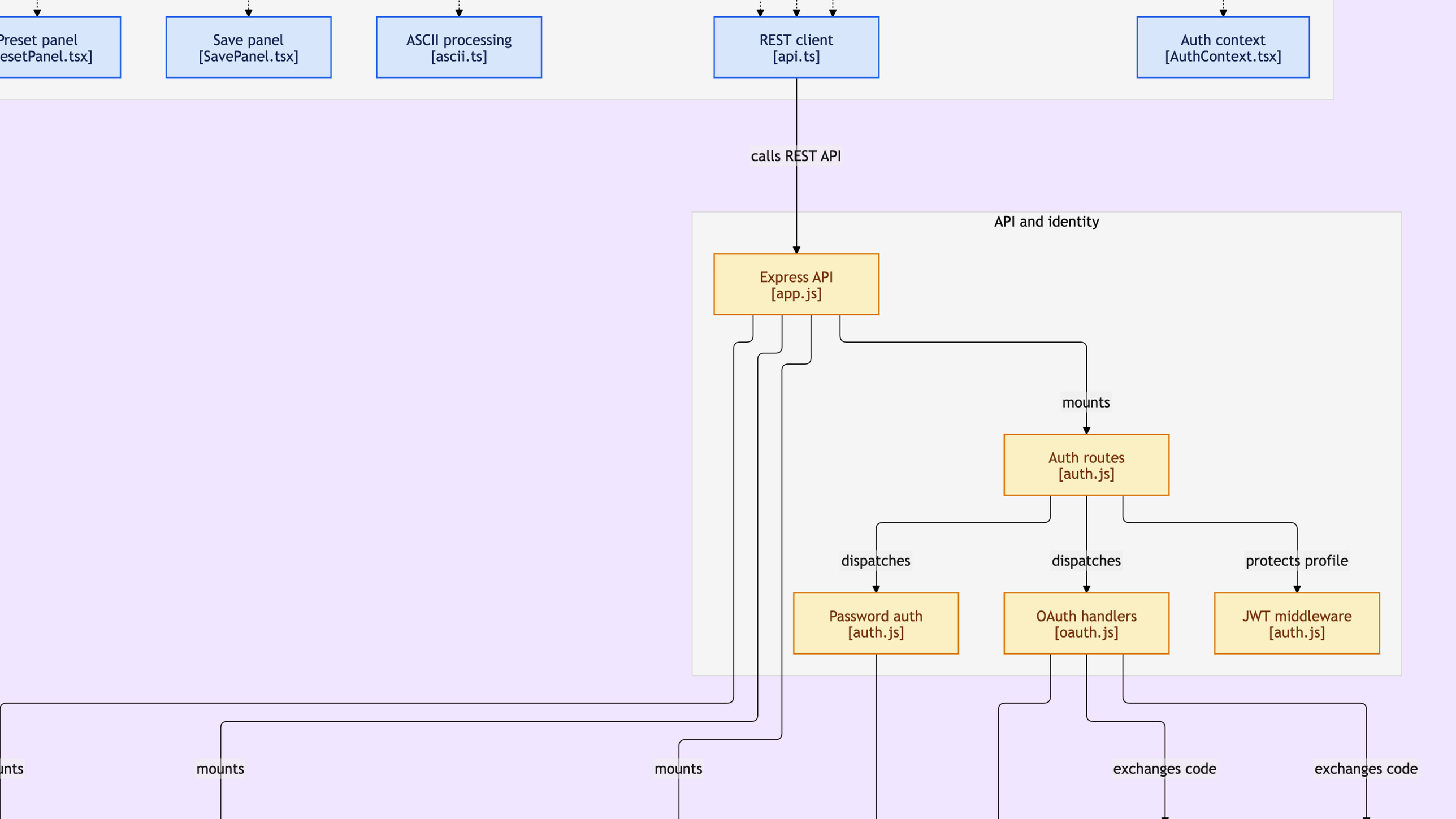
1. Image to ASCII: Convert images into ASCII art.
2. Video to ASCII: Transform video frames into ASCII animations.
3. Customizable Output: Adjust character density and resolution.
4. Live Preview: View conversions instantly.
5. Download & Export: Save the generated ASCII art.
6. Simple Interface: Easy-to-use web application.

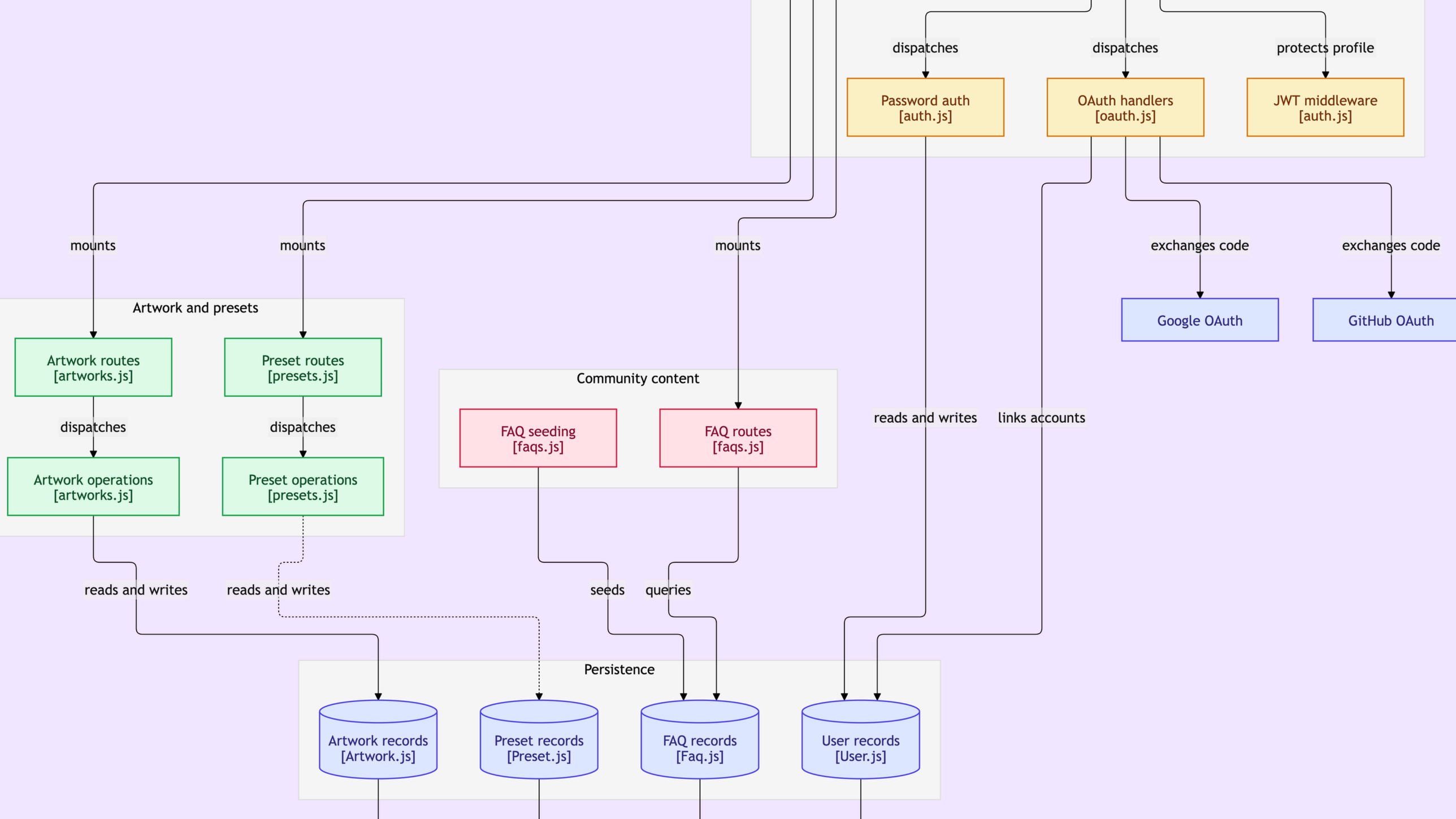


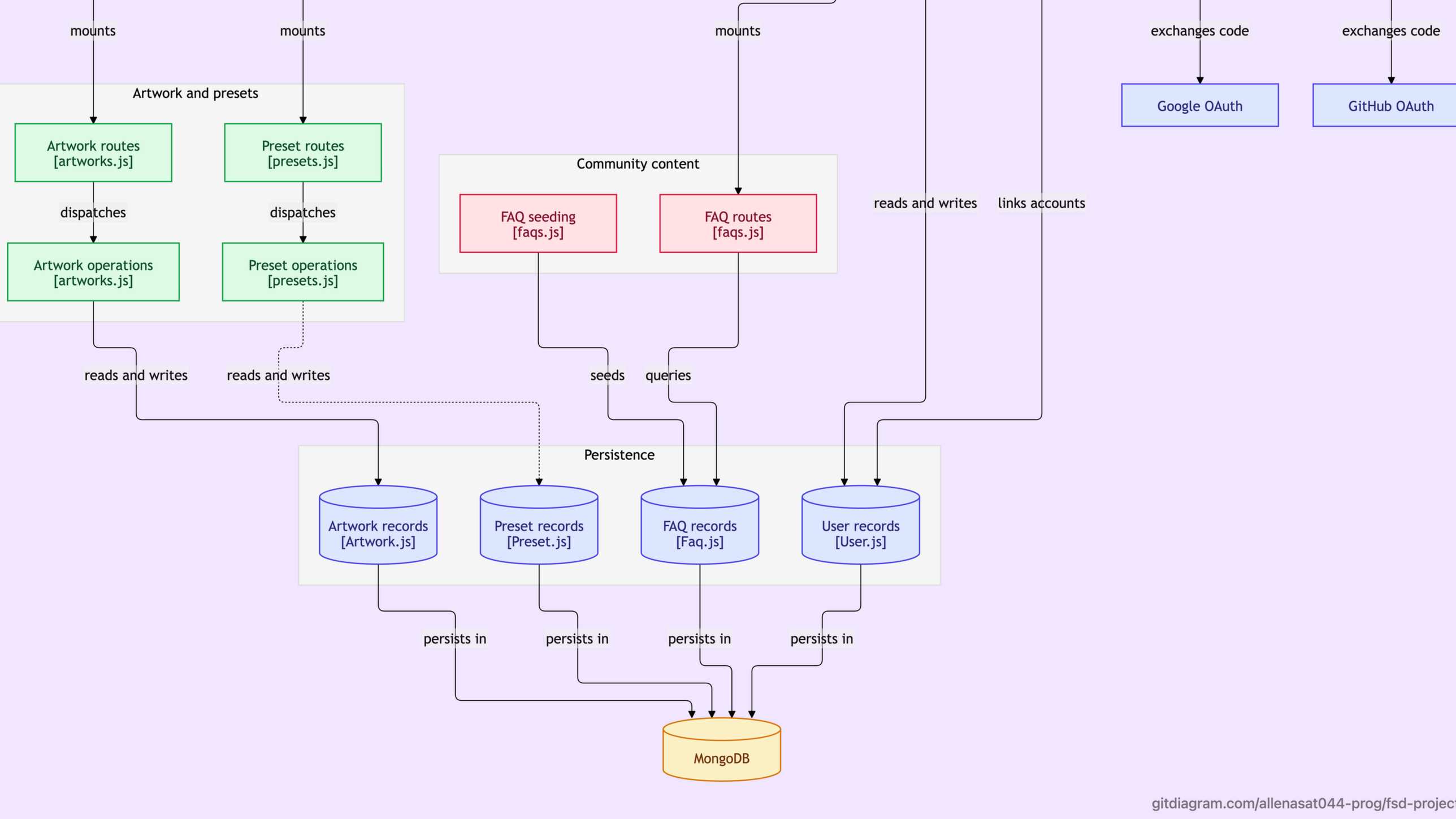


# SYSTEM ARCHITECTURE


Visitor
uses
Client experience
React app
[App.tsx]
routes to
routes to
routes to
routes to
routes to
ASCII converter
[AsciiDashboard.tsx]
Artwork detail
[ArtworkPage.tsx]
Sign-in pages
[AuthPage.tsx]
Gallery
[Gallery.tsx]
OAuth callback
[AuthCallback.tsx]
shows
shows
converts input
sends requests
submits credentials
loads gallery
sets session
Preset panel
[PresetPanel.tsx]
Save panel
[SavePanel.tsx]
ASCII processing
[ascii.ts]
REST client
[api.ts]
Auth context
[AuthContext.tsx]
calls REST API
API and identity


Preset panel
esetPanel.tsx]
Save panel
[SavePanel.tsx]
ASCII processing
[ascii.ts]
REST client
[api.ts]
Auth context
[AuthContext.tsx]
calls REST API
API and identity
Express API
[app.js]
mounts
Auth routes
[auth.js]
dispatches
dispatches
protects profile
Password auth
[auth.js]
OAuth handlers
[oauth.js]
JWT middleware
[auth.js]
unts
mounts
mounts
exchanges code
exchanges code


dispatches
dispatches
protects profile
Password auth
[auth.js]
OAuth handlers
[oauth.js]
JWT middleware
[auth.js]
mounts
mounts
mounts
exchanges code
exchanges code
Google OAuth
GitHub OAuth
Artwork and presets
Artwork routes
[artworks.js]
Preset routes
[presets.js]
Community content
FAQ seeding
[faqs.js]
FAQ routes
[faqs.js]
reads and writes
links accounts
dispatches
dispatches
Artwork operations
[artworks.js]
Preset operations
[presets.js]
reads and writes
reads and writes
seeds
queries
Persistence
Artwork records
[Artwork.js]
Preset records
[Preset.js]
FAQ records
[Faq.js]
User records
[User.js]




gitdiagram.com/allenasat044-prog/fsd-project

# ALERTING AND REPORTING

1. Image & Video Conversion: Convert media into ASCII art.
2. Customization: Adjust characters, resolution, and output appearance.
3. Real-Time Preview: View results instantly.
4. Download & Export: Save generated ASCII art.
5. Reporting: Generate conversion reports and track activity.
6. MongoDB Integration: Store user data, conversion history, and reports.
7. Responsive UI: Simple and user-friendly interface.

Drop an image or click to choose

Resolution (columns) 500

Contrast 1.55

Brightness 0.1

Character set

Detailed (70 levels)

Mode

True color

Invert Mirror

Download format

png

Download

**Save to gallery**

sample

Tags, comma separated (max 5)

Public

Save

**My presets**

No presets yet

Apply Delete

New preset name Save





# Gallery

ASCII art saved by the community, and your own creations.

Public My creations Liked

All sources Camera Video Image

Search titles Search

Newest first

1 piece



**cat**

Shashank Patel · 332×166 · 26/10/2026

#cute

♡ 0 View



Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

- 127.0.0.1:27017
  - admin
  - ascii_cam
    - artworks
    - faqs
    - presets
    - users
  - config
  - local

artworks

127.0.0.1:27017 > ascii_cam > artworks

Open MongoDB shell

Documents 1 | Aggregations | Schema | Indexes 6 | Validation

Type a query: { field: 'value' } or Generate query

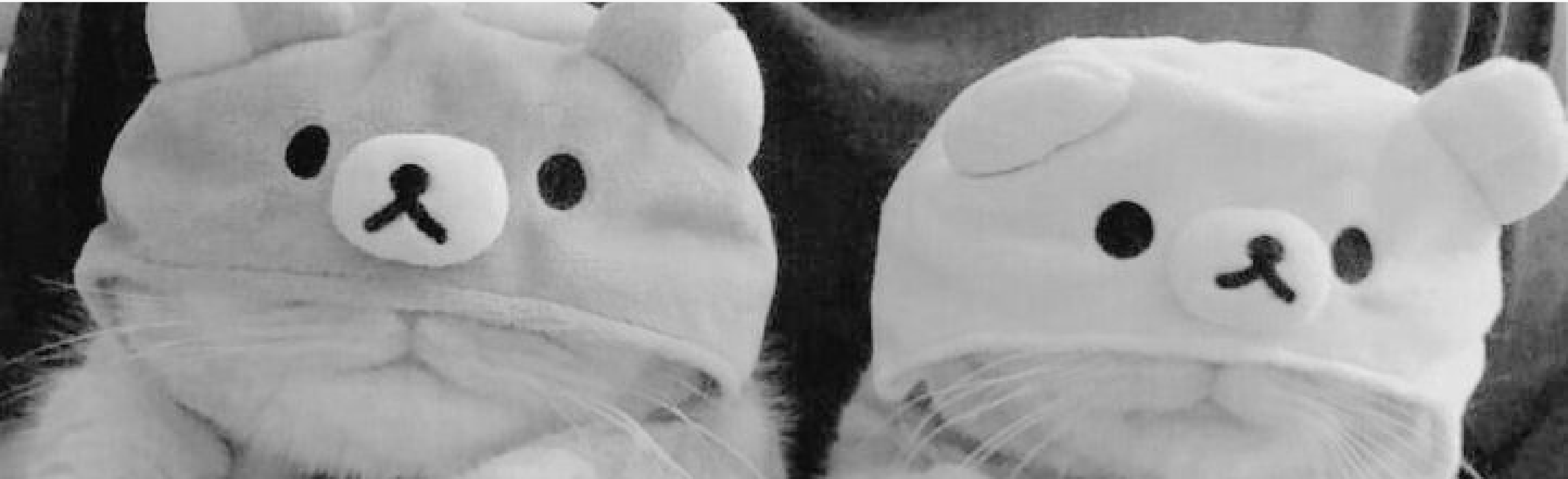
Explain | Reset | Find | Options

ADD DATA | BULK | EXPORT DATA | EXPORT CODE

25 | 1 – 1 of 1


```
_id: ObjectId('6adf5ddc5f3912e278e75f21')
user: ObjectId('6adf5da15f3912e278e75f10')
title: "cat"
text: "@@@
```

# TESTING AND VALIDATION





# APPLICATIONS AND FUTURE SCOPE

Applications

- Linux servers and virtual machines.
- Development environments and storage servers.
- Backup systems and data-processing machines.
- System administration labs.

Future enhancements

- Email and mobile notifications.
- Authentication and role-based access.
- Multi-server monitoring from one dashboard.
- Improved forecasting and anomaly detection.
- Configurable retention and historical trend visualizations.



# THANK YOU

